

Engineering Programming II: Possible Topics

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1 Intro to Python

1.1 IDEs

- i. Anaconda 3
- ii. Spyder IDE and Jupyter Notebooks/Lab

1.2 Language and syntax

- i. Interpreted vs. compiled languages
- ii. Tabbing
- iii. Types
 - A. type()
 - B. Numeric: int, float
 - C. Boolean
 - D. String
 - E. Casting
 - F. Dynamic typing

iv. Lists

- A. Slicing
- B. Conditional indexing
- C. 2D lists
- D. Lists vs. numpy arrays

v. Loops

- A. For loops
- B. in
- C. While loops
- D. break/continue

vi. Conditionals: if, elif, else

vii. User-defined functions

- A. def
- B. return
- C. Positional/keyword arguments

viii. Importing packages

- ix. File I/O (standard, numpy, pickle)

x. Error messages

1.3 Fundamental packages

i. Math

ii. NumPy

A. math

B. arrays: sorting, argmax, where, argwhere, ones, zeros, eye, rand, randn, identity, cumsum, all, any, isinf, isnan,

C. histogram, bincount

D. stats: mean, mediand, std,

E. random

F. linalg

G. fft

iii. Matplotlib

A. plot, hist, imshow, scatter

B. fig plt ax

C. subplots

iv. SciPy

A. odes

B. optimization

2 Computer arithmetic

2.1 Data types

2.2 Floating point representation

2.3 Integer division

2.4 Error

3 Random variables, RNG, and Statistics

3.1 Probability distributions

3.2 Law of Large Numbers (std of mean, integral of f over p)

3.3 Random number generation

3.4 numpy.random

3.5 Random walk

3.6 Drawing from a custom distribution (linear interpolation/binning)	ii. Heun's method (2nd order)
	iii. Runga-Kutta (4th order)
	iv. SciPy ODE solvers
4 Monte Carlo simulation	10.3 Examples
4.1 Calculating π	i. Exponential growth
4.2 Calculating integrals	ii. Projectile motion
4.3 Functions of distributions	iii. Lotka-Volterra predator-prey
	iv. Simple pendulum
	v. n -body problem
5 Linear Algebra (solving linear systems)	11 PDEs (time-dependent)
5.1 numpy.linalg	11.1 Discretization
5.2 Condition number	11.2 Methods
	i. Forward difference
	ii. Backward difference
6 Root-finding (solving nonlinear systems)	iii. Crank-Nicholson
6.1 Newton's method	iv. Spectral
7 Regression	11.3 Examples
7.1 Pearson correlation coefficient	i. Advection
7.2 Linear Least squares	ii. Wave equation
	iii. Schrodinger equation
8 Optimization	12 Agent-based modeling
8.1 Calculus review: $df/dx = 0$	12.1 Cellular automata
8.2 Gradient descent	12.2 Conway's game of life
8.3 Simulated annealing	12.3 SIR pandemic modeling
8.4 Stochastic gradient descent	
8.5 Genetic algorithms	13 Markov chains
9 Fourier Analysis	13.1 Simple example
10 ODEs (IVPs)	13.2 Definition
10.1 Discretization	13.3 NLP n -grams
10.2 Methods	13.4 Markov chain Monte Carlo methods
i. Euler's method (1st order)	